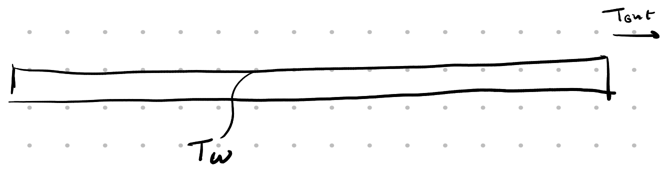


convection

↑↓

cond

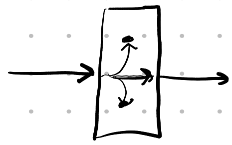
Rayleigh Number (Ra) = Graychoffs Number (Gr) × Prandtl Number (Pr)

$$\frac{\text{buoyancy force}}{\text{thermal diffusivity}} = \frac{\text{buoyancy force}}{\text{momentum diffusivity (viscous force)}} \times \frac{\text{momentum diffusivity (viscous force)}}{\text{thermal diffusivity}}$$

$$\alpha = \frac{Q_{cond}}{Q_{stored}} \ (m^2/s)$$

Buoyancy

Diffusivity



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mean bulk temperature

$$T_{mb} = \frac{T_{in} + T_{out}}{2} = T_\infty$$

film temperature (average temperature)

$$T_f = \frac{T_{mb} + T_w}{2}$$

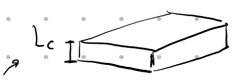
T_w – surface temp

expansion co-efficient

$$\beta = \frac{1}{T_f} \rightarrow K$$

$$R_a = Gr\,Pr = \frac{(9.81)\beta(T_w - T_\infty)L_c^3}{\nu^2} Pr$$

$$Nu = \frac{hL}{k}$$



Geometry → L_c → R_a → Nu → h

$$\dot{Q}_{conv} = hA(T_w - T_\infty)$$

Fins

$$Nu = \frac{S h}{k} \qquad S_{opt} = f(Ra, L)$$

$$A = 2 N_{fins} L H$$

$$N_{fins} = \frac{w}{t + s}$$

$$L = L_c$$